

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

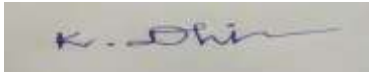
Total Marks: 50

Code of Conduct

I K.Dhilip/192365013 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-	Covers most expected points	Covers only basic points	Very limited coverage

		points completely			
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition

Cloud computing is a technology that allows people and organizations to use computing services such as servers, storage, databases, networking, and software through the internet instead of owning and maintaining physical computers or servers. It helps users access resources whenever they need them while paying only for what they use.

Characteristics of Cloud Computing

1. On-Demand Self-Service

One of the main features of cloud computing is that users can access computing resources whenever they need them without waiting for assistance from the service provider. For example, if a company needs additional storage or a new virtual machine, it can create it within a few minutes through an online portal. This saves both time and effort.

2. Broad Network Access

Cloud services can be accessed from anywhere using an internet connection. Users can log in through laptops, desktop computers, smartphones, or tablets. This makes it easy for employees and students to work from home, college, or any other location without carrying important files with them.

3. Resource Pooling

Cloud service providers maintain a large pool of shared computing resources that are used by many customers at the same time. Even though the hardware is shared, each user's data remains private and secure. This sharing of resources helps reduce costs and improves the efficient use of infrastructure.

4. Rapid Elasticity

Cloud computing allows users to increase or decrease computing resources based on their needs. For example, during a festive sale, an online shopping website may receive millions of visitors. The cloud can automatically provide additional servers to handle the increased traffic. Once the traffic decreases, the extra resources are removed, helping the company avoid unnecessary expenses.

How Cloud Computing is Different from Traditional On-Premise Computing

Cloud Computing	Traditional On-Premise Computing
Resources are available through the internet whenever required.	Resources are installed and managed within the organization.
Users can easily increase or decrease resources based on demand.	Expanding resources requires purchasing and installing new hardware.
Customers pay only for the resources they use.	Organizations must invest a large amount of money in advance for hardware and maintenance.

Maintenance and updates are handled by the cloud provider.	The organization's IT team is responsible for maintenance and updates.
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Conclusion

Cloud computing has transformed the way organizations use technology by providing flexible, scalable, and cost-effective computing services over the internet. Features such as on-demand self-service, broad network access, resource pooling, and rapid elasticity make it more efficient and convenient than traditional on-premise computing. As a result, many businesses, educational institutions, and individuals now prefer cloud computing because it reduces costs, improves accessibility, and supports business growth.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Cloud computing did not appear overnight. It developed gradually as computer technology evolved over several decades. The journey started with large physical computers, moved through networking and virtualization, and finally reached internet-based software and cloud services. Each stage solved the limitations of the previous one and contributed to the development of modern cloud platforms.

1. Hardware Evolution (1950s–1970s)

The early stage of computing was dominated by large mainframe computers. These machines were expensive, occupied large spaces, and were owned only by governments, universities, and large organizations. Multiple users shared the same computer through terminals, introducing the concept of resource sharing. Although computing power was limited, this stage laid the foundation for future cloud technologies.

2. Personal Computer (PC) Revolution (1980s)

The introduction of personal computers made computing affordable and accessible to individuals and small businesses. People could now own their own computers and run applications locally. However, software, storage, and maintenance had to be managed individually, making upgrades and data sharing difficult.

3. Networking and Client-Server Computing (1990s)

With the growth of computer networks and the internet, organizations adopted the client-server model. In this model, servers stored data and applications, while client computers accessed them over a network. This improved collaboration and centralized data management, but organizations still needed to purchase, maintain, and upgrade their own servers.

4. Virtualization Technology (Late 1990s–2000s)

Virtualization introduced the ability to run multiple virtual machines on a single physical server. This greatly improved hardware utilization, reduced costs, and allowed resources to be allocated dynamically. Virtualization became one of the most important technologies behind cloud computing because it enabled flexible and efficient resource management.

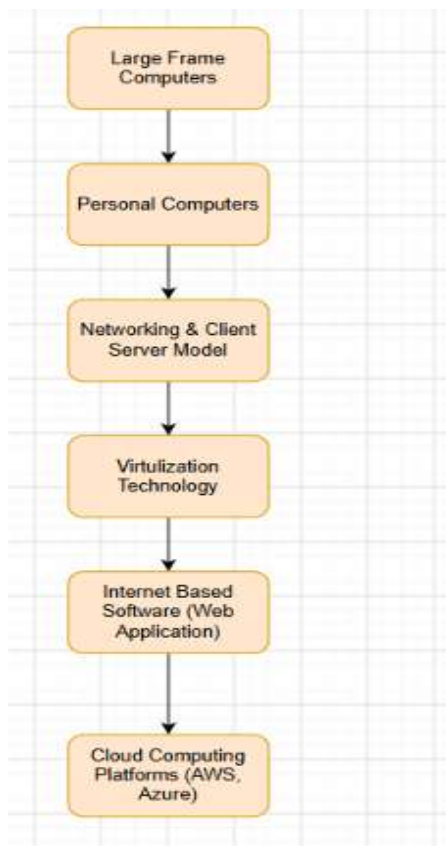
5. Internet Software Evolution (2000s)

As internet speed and reliability improved, software applications began to move from local computers to web browsers. Instead of installing software on every device, users could access applications through the internet. Services such as web-based email, online document editing, and file storage became popular. This gave rise to the Software as a Service (SaaS) model and changed how software was delivered.

6. Modern Cloud Platforms (2010s–Present)

The combination of virtualization, high-speed internet, distributed data centers, and web-based software led to modern cloud computing platforms. Today, providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform offer computing resources, storage, databases, networking, and artificial intelligence services on demand. Users can scale resources instantly, pay only for what they use, and access services from anywhere in the world.

Evolution Flow Diagram



How These Changes Enabled Modern Cloud Platforms

- **Hardware evolution** introduced the idea of sharing computing resources.
- **Personal computers** made technology accessible to more users.
- **Networking** enabled communication and centralized data storage.
- **Virtualization** improved hardware efficiency and allowed multiple virtual servers to run on one physical machine.

- **Internet software** allowed applications to be accessed through web browsers instead of being installed locally.
- **Cloud platforms** combined all these technologies to provide scalable, reliable, secure, and cost-effective computing services over the internet.

Conclusion

The evolution of cloud computing has been a gradual process driven by advancements in hardware, networking, virtualization, and internet-based software. These developments transformed traditional computing into today's cloud platforms, allowing users to access powerful computing resources anytime and anywhere without investing in expensive infrastructure. This has made cloud computing an essential technology for businesses, educational institutions, and individuals worldwide.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization in Cloud Computing

Server virtualization is one of the most important technologies behind cloud computing. It allows a single physical server to be divided into multiple virtual servers, known as Virtual Machines (VMs). Each virtual machine has its own operating system, applications, storage, and resources, even though all of them share the same physical hardware.

Virtualization helps cloud providers make better use of their hardware resources. Instead of keeping one application on one physical server, several virtual machines can run on the same server. This reduces hardware costs, saves energy, and improves system performance.

Another important benefit is flexibility. Virtual machines can be created, modified, or removed within minutes according to user requirements. If one virtual machine fails, the others continue to operate independently, improving the reliability and availability of cloud services.

Server virtualization also supports workload balancing and disaster recovery. Cloud providers can easily move virtual machines from one physical server to another without affecting users. This makes cloud services more reliable and ensures minimal downtime.

Benefits of Server Virtualization

- Improves hardware utilization by running multiple virtual machines on one server.
- Reduces infrastructure and maintenance costs.
- Enables quick creation and deployment of new servers.
- Supports scalability based on user demand.
- Improves availability, backup, and disaster recovery.
- Forms the foundation for Infrastructure as a Service (IaaS) in cloud computing.

Difference Between Virtualization and Cloud Computing

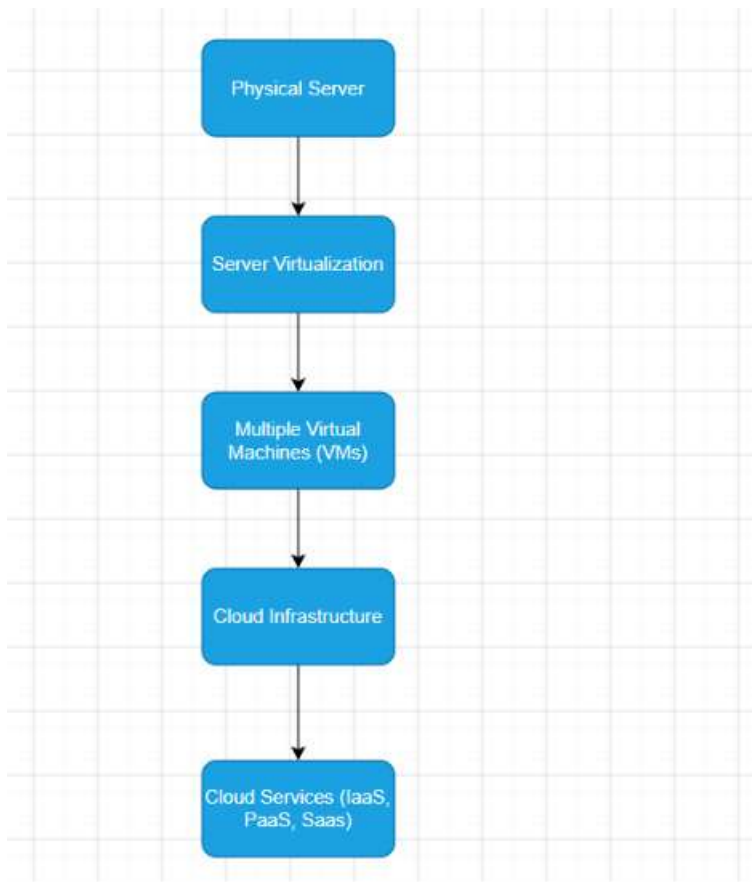
Virtualization (Enabling Technology)	Cloud Computing (Service Model)
Virtualization is a technology that creates multiple virtual machines on a single physical	Cloud computing is a model that delivers computing services over the internet.

server	
Its main purpose is to improve hardware utilization and resource management.	Its main purpose is to provide computing resources as on-demand services.
It works at the infrastructure level.	It works at the service delivery level.
It can exist without cloud computing.	Cloud computing depends heavily on virtualization to function efficiently.
Managed mainly by system administrators.	Used by businesses and individuals through cloud service providers.
Example: VMware, Microsoft Hyper-V, Oracle VirtualBox	Example: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP).

Relationship Between Virtualization and Cloud Computing

Virtualization is the foundation of cloud computing, while cloud computing is the service built on top of virtualization. Virtualization makes it possible to share physical resources efficiently, and cloud computing uses these virtualized resources to provide services such as virtual machines, storage, databases, and applications through the internet.

Simple Flow Diagram



Conclusion

Server virtualization plays a vital role in cloud computing by allowing multiple virtual machines to run on a single physical server, improving resource utilization, reducing costs, and enabling scalability. While virtualization is the technology that creates and manages virtual resources, cloud computing is the

service model that delivers those resources to users over the internet. Together, they form the foundation of modern cloud platforms and make cloud services flexible, reliable, and cost-effective.

4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Web services allow different applications to communicate with each other over a network. The two most commonly used web service architectures are **SOAP (Simple Object Access Protocol)** and **REST (Representational State Transfer)**. Although both are used to exchange information between applications, they differ in their communication style, data format, performance, and suitability for cloud computing.

SOAP-Based Web Services

SOAP is a protocol that follows strict standards for communication between applications. It mainly uses XML to exchange data and provides built-in security, reliability, and error handling. SOAP is commonly used in banking, healthcare, and enterprise applications where secure and reliable communication is essential.

REST-Based Web Services

REST is an architectural style that uses standard HTTP methods such as GET, POST, PUT, and DELETE to communicate between applications. It supports multiple data formats, including JSON, XML, HTML, and plain text. REST is lightweight, faster, and easier to develop, making it the preferred choice for modern cloud and web applications.

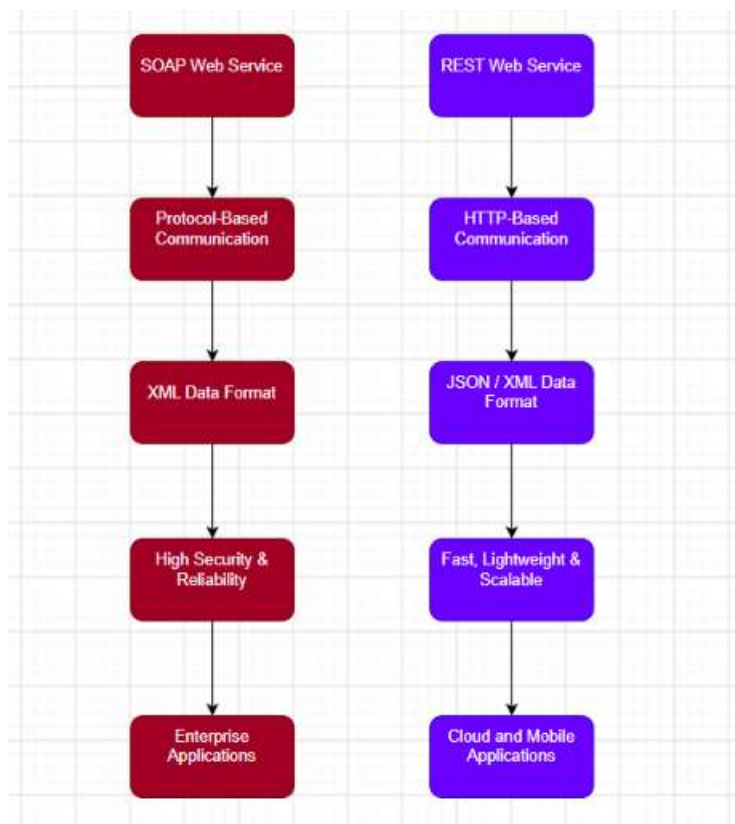
Feature	SOAP-Based Web Services	REST-Based Web Services
Communication Style	Uses a protocol with strict messaging standards	Uses an architectural style based on HTTP methods.
Data Exchange	Primarily uses XML for sending and receiving data.	Supports JSON, XML, HTML, and plain text, with JSON being the most common.
Performance	Slower because XML messages are larger and require more processing.	Faster because JSON is lightweight and requires less bandwidth.
Security	Provides built-in security features such as WS-Security.	Relies mainly on HTTPS, OAuth, JWT, and other web security mechanisms.
Flexibility	Less flexible due to strict standards and message formats.	Highly flexible and easy to integrate with different applications.
Ease of Development	More complex to develop and maintain.	Simpler and easier to develop and use.
Suitability for Cloud Applications	Suitable for enterprise applications requiring high security and reliability	Best suited for cloud computing, mobile applications, and web APIs because it is lightweight and scalable

Why REST is Preferred for Cloud Applications

REST has become the preferred choice for cloud computing because it is lightweight, fast, and easy to scale. It uses JSON, which reduces the amount of data transferred over the network and improves performance. REST APIs can also be accessed easily from web browsers, mobile applications, and IoT devices, making them ideal for cloud-based services.

SOAP is still important for applications where security, reliability, and transaction support are critical. However, for most modern cloud platforms, REST is preferred because it offers better performance and easier integration.

Simple Comparison Diagram



Conclusion

SOAP and REST are both important web service technologies, but they are designed for different purposes. **SOAP** is a protocol that provides strong security, reliability, and standardized communication, making it suitable for enterprise systems. **REST** is a lightweight architectural style that supports multiple data formats, especially JSON, making it faster, more flexible, and highly suitable for modern cloud applications. As a result, most cloud platforms and web services today prefer REST because of its simplicity, scalability, and better performance.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Cloud computing provides different service models based on the needs of users and organizations. The four common models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Anything as a Service (XaaS). Each model offers a different level of control, responsibility, and functionality.

1. Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) provides virtualized computing resources such as servers, storage, networking, and virtual machines over the internet. The cloud provider manages the physical infrastructure, while the user is responsible for installing and managing the operating system, applications, and data.

Real-World Example: Amazon EC2 (AWS) allows users to create virtual servers and use them according to their requirements without purchasing physical hardware.

2. Platform as a Service (PaaS)

Platform as a Service (PaaS) provides a complete platform for developing, testing, deploying, and managing applications. The cloud provider manages the infrastructure, operating system, and runtime environment, allowing developers to focus only on writing and deploying code.

Real-World Example: Google App Engine enables developers to build and deploy applications without worrying about server management.

3. Software as a Service (SaaS)

Software as a Service (SaaS) delivers ready-to-use software applications through the internet. Users simply access the application using a web browser without installing or maintaining the software. The cloud provider manages everything, including updates, security, and maintenance.

Real-World Example: Google Docs allows users to create, edit, and share documents online without installing any office software.

4. Anything as a Service (XaaS)

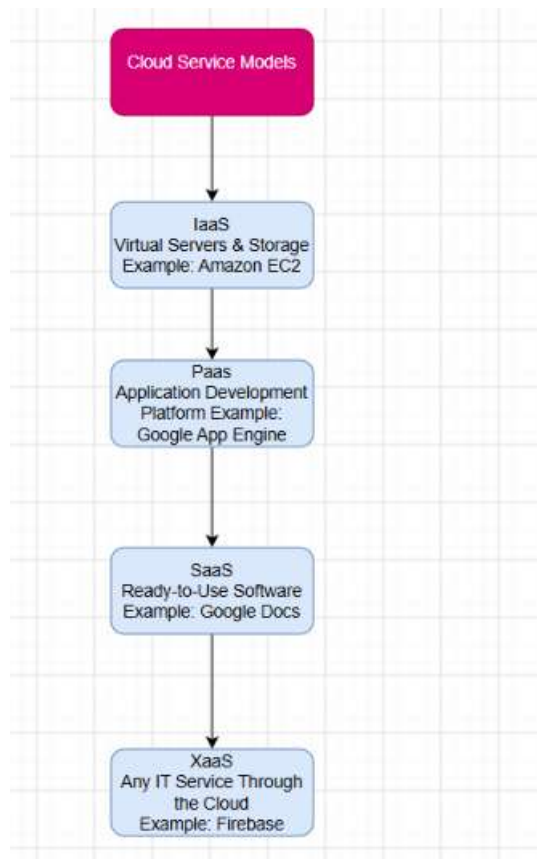
Anything as a Service (XaaS) is a broad term that refers to delivering almost any IT service through the cloud. It includes services such as Database as a Service (DBaaS), Storage as a Service (STaaS), Security as a Service (SECaaS), Function as a Service (FaaS), and many others.

Real-World Example: Firebase provides multiple cloud services such as databases, authentication, cloud storage, and hosting, making it an example of XaaS.

Service Mode	What It Provides	Who Manages Most Resources?	Real-World Example
IaaS	Virtual servers, storage, and networking	Cloud provider manages hardware; user manages OS and applications	Amazon EC2 (AWS)

PaaS	Platform for developing and deploying applications	Cloud provider manages infrastructure and platform	Google App Engine
SaaS	Ready-to-use software applications	Cloud provider manages everything	Google Docs
XaaS	Any IT service delivered through the cloud	Depends on the type of service provided	FireBase

Simple Wroflow Diagram



Conclusion

The different cloud service models are designed to meet different user requirements. IaaS provides virtual infrastructure, PaaS offers a platform for application development, SaaS delivers ready-to-use software, and XaaS covers a wide range of cloud-based services. These models help organizations reduce costs, improve flexibility, and access technology without investing in expensive hardware and software infrastructure.